TLS PROPOSAL

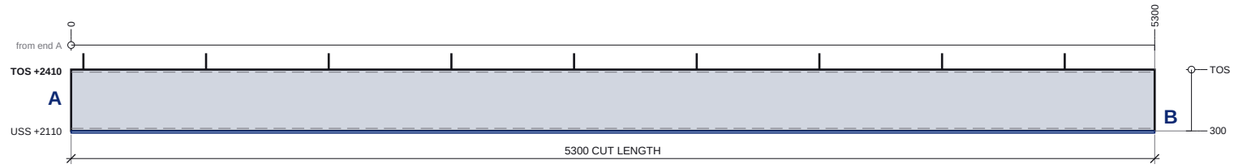
NOT FOR FABRICATION

FOR ENGINEERS REVIEW

until the engineer has reviewed + signed off

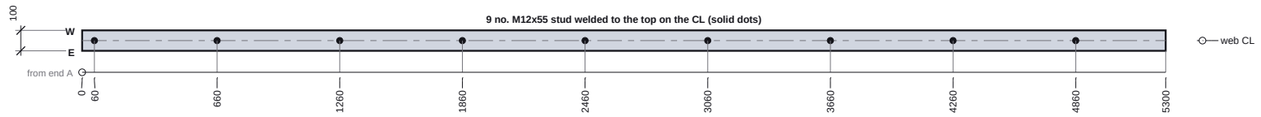
SB4 ELEVATION

Scale 1:30 | viewed from the E side, end A (S) left



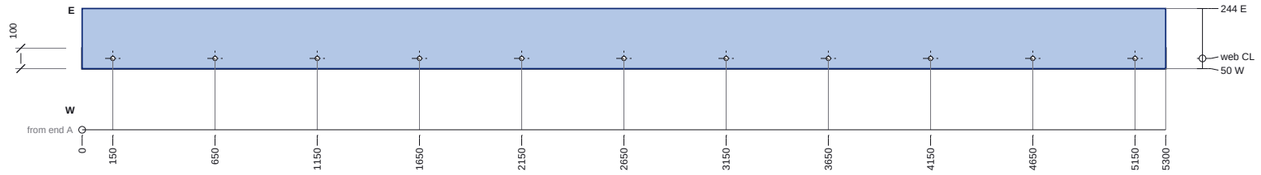
SB4 TOP - PLAN

1:30



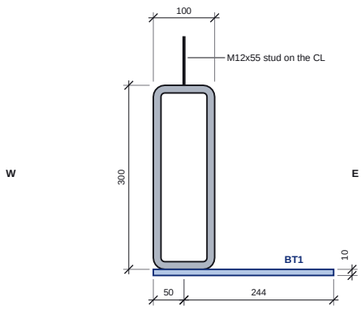
SB4 UNDERSIDE - VIEWED FROM BELOW

1:30



SECTION THROUGH SB4 1:10

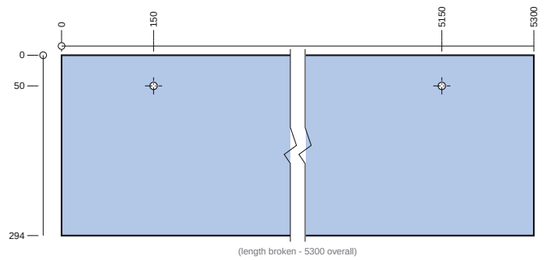
looking from end A to end B: W left, E right



BT1: 50 W / 244 E of the CL, underside 310 below TOS

Knee connections + assembly: see P4-ST-GP1

BT1 PLATE DETAIL 1:10



u along the beam from the south end (left edge), v from the west edge (top edge)

5300 x 294 x 10 thk S275
11 no. Ø12 holes
122.22 kg qty 1 on SB4
Hole u (all): 150, 650, 1150, 1650, 2150, 2650, 3150, 3650, 4150, 4650, 5150 (middle ones in the broken length)
Weld: 6mm FW both toes to the RHS (flare-bevel at the RHS corners, throat 5 min); ends 6 FW to the post inner faces
boot plate under the RHS, 10 back from the stone face; Ø12 vents in the 100 lap under the RHS (sealed lap, BS EN ISO 14713-2), plugged after galvanising - galvaniser to confirm
TLS detail - engineer to confirm

HOLE SCHEDULE

No.	FACE	X FROM A	POSITION	HOLE / STUD	PURPOSE
S1	top	60	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S2	top	660	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S3	top	1260	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S4	top	1860	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S5	top	2460	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S6	top	3060	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S7	top	3660	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S8	top	4260	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
S9	top	4860	on CL	stud M12x55	M12x55 stud welded to the top (6 FW all round), not a hole [TLS detail - engineer to confirm]
BT1 V1..V11	BT1	150..5150 @500	u / v: BT1 PLATE DETAIL	Ø12	11 no. vent / drain holes in BT1 - plug / seal after galvanising

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
SB4	RHS300x100x12.5	S355J2H	5300	1	383.2
BT1	PL 5300x294x10	S275		1	122.22
TOTAL					**505.4 kg**

CONNECTIONS

Detail GP1 knee: SB4 end A/B on PC1/PC2 - Shop-welded knee (R4): RHS square-cut ends welded all round to the post inner faces, 8 FW; BT1 ends 6 FW to the post faces. Frame galvanised as one piece (galvaniser to confirm the bath + dip; temporary bolted spreader between the bases until set; post centres +/-3 after galvanising). FOR ENGINEERS REVIEW

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. RHS S355J2H (EN 10210-1), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Holes / plates marked TLS detail are not on the engineer's drawings: TLS detail - engineer to confirm before fabrication.
8. GP1 - TLS PROPOSAL R4 FOR ENGINEERS REVIEW (replaces S-0010 UB203x133x30); calc on the GP1 sheet
9. RHS side walls land on the SHS corner radius (flat +/-45): flare-bevel weld, throat 5 min - engineer to confirm.
10. Studs welded before galvanising: chase the threads / oversize-tapped HDG nuts (BS EN ISO 10684).
11. GP1 R4 - PRELIMINARY, FOR ENGINEER REVIEW (TLS proposal, not the engineer's design). Replaces S-0010's UB203x133x30: carries the 5300 door opening AND ~0.5 m of stone (main-house stone eaves).
12. LOADS (char.): trusses S-0012 DL 15 / IL 10 kN each end -> 5.4 / 3.6 kN/m; stone 0.5x0.1x23 = 1.15 kN/m; boot + timber + self weight 1.0 kN/m; door rails + operator 2.0 kN mid-span; wind 12 kN SLS / 18 kN ULS at the eaves.
13. RHS 300x100x12.5: I 9103 cm4, Wpl 806 cm3. ULS M 61 kNm vs Mpl 286 kNm. Deflection (SLS incl. door, pinned, L 5420) 6.9 mm = L/787 < 8 mm (client limit).
14. SHS 120x120x10 to TOS: N 44 kN + knee M 29.7 kNm vs Mpl 62 kNm; sway 6.8 mm = h/326.
15. KNEE (R4): RHS ends welded all round to the post inner faces, 8 FW: couple 29.7 / 0.288 = 103 kN per RHS wall vs 136 kN; post face (b1/b0 0.83) ~390 kN; torsion from the boot 0.84 kNm per knee.
16. BASE: 400x200x15, 54 clear of the stone leaf; N 44 kN, H 9 kN per base (ULS); pinned base assumed for sway. Boot toe ~0.22 kNm/m -> 13 MPa in the 10 plate.
17. FOR THE ENGINEER: loads, wind share, knee welds + post face (stiffeners in the SHS?), base fixity, padstones on the garage return foundations (min 500 x 300), cavity tray / thermal bridge at the boot, 30 min fire protection (S-0011 9.02).

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

SB4 RHS300X100X12.5 FABRICATION

GP1 portal beam between the posts over the 5300 garage door; carries GT1 + ~0.5 m stone on the boot

SCALE	SIZE	DATE	DRAWN
1:30 / details as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-SB4	C03

Do not scale. All dimensions in mm. Check all dimensions on site.